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Stress detection in daily life scenarios using smart phones and wearable sensors: A survey

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ABSTRACT

Stress has become a significant cause for many diseases in the modern society. Recently, smartphones, smartwatches and smart wrist-bands have become an integral part of our lives and have reached a widespread usage. This raised the question of whether we can detect and prevent stress with smartphones and wearable sensors. In this survey, we will examine the recent works on stress detection in daily life which are using smartphones and wearable devices. Although there are a number of works related to stress detection in controlled laboratory conditions, the number of studies examining stress detection in daily life is limited. We will divide and investigate the works according to used physiological modality and their targeted environment such as office, campus, car and unrestricted daily life conditions. We will also discuss promising techniques, alleviation methods and research challenges.

1. Introduction

In the last two decades, researchers have realized that there is an important relationship between the physical health of an individual and his/her emotional state [1]. This has led to increasing interest in affective computing (AC) which makes use of technology to recognize the affective state of a person. Dr. Rosalind Picard of the Massachusetts Institute of Technology (MIT) published the first book on affective computing [2]. It has since become a prominent branch under human-computer interaction area [3,4]. It makes use of physiological and physical manifestations of an individual to detect the current emotion. Prominent elementary emotional states that can be inferred in affective computing are joy, anger, surprise, disgust, sadness, and fear [1]. Stress has been recently added to abovementioned emotions that can be determined. It can be defined as the reaction that people may have when they are subject to demands and pressures which do not correspond to their knowledge and abilities and that can challenge their handling capabilities [5]. Stress is a significant problem of the modern society. It is a growing issue and it has become an inescapable part of our daily lives. The early detection will decrease the damage it costs and prevents it from being chronic. The damages of stress on human health has been known by the researchers and a significant amount of efforts have been made recently to develop an automatic stress measuring system by making use of smart devices and advanced AC algorithms. Possible application areas of automatic stress detection systems can be listed as driver stress monitoring in automobile environments, passenger stress detection and alleviation in the commercial flights in airplanes especially for passengers with flying phobia, stress assessment of workers to augment efficiency in the workplaces (factories and offices), stress handling evaluation for applicants in job interviews and supporting psychologists in online therapy sessions by continuously monitoring stress level of patients.

There are some surveys in this area. In [6], authors described the types of physiological signals without mentioning the related research and papers. They only provided the types of physiological signals and some features of them. Thapliyal et al. [7] introduced some devices in the market for stress detection. Greene et al. [8] provided physiological signals and commonly used devices for each type of signal with a brief literature survey. Gulhane et al. [9] also presented a very short survey that mentioned a few techniques in the literature. The disadvantage of these works is that although they explained the technologies in detail and off-the-shelf products for these technologies, they did not provide the literature which are using these technologies. Another drawback of these articles is the lack of literature using smartphones, smartwatches and smart bands. In [10,11], authors described research which detects stress at workplaces and offices. However, the recent direction of the research for stress detection is to recognize stress not only in the workplaces but also in the daily lives of individuals. To the best of our

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knowledge, our survey is the first one that reviews the stress detection research in all environments including unrestricted daily life environments. Our paper investigates four significant novel aspects of stress detection research:

- The types of devices and commonly extracted features were discussed for each physiological signal.
- The stress detection research is divided into laboratory, office, school, automobile and daily life environments and research insights were provided for each environment.
- The ultimate aim of stress detection studies is the relaxation of an individual after detecting stress. We included scientifically proven relaxation apps and methods that researchers can apply to this end.
- Research directions in the stress detection topic and open research problems were identified.

The preliminary works for detecting stress have started in controlled laboratory environments. However, scientists realized that the stress level experienced in the laboratory is different compared with the stress in daily life [1]. Another important reason for this trend shift in stress recognition studies was that users do not want invasive measurement techniques because they are not comfortable with them. For these reasons, noninvasive stress measurements in daily life have become a hot topic for researchers. Since smartphones and wearable devices are these kinds of instruments and used every day, they are chosen for daily stress detection studies.

In this survey, we first provide the definition and impacts of stress in Section 2 and Section 3 respectively. We further briefly mention the stress induction methods in Section 4. Physiological signals that are used to detect stress, widely used devices for data collection and features that are used commonly for each signal are given in Section 5. Real-life data collection challenges for stress detection research are presented in Section 6. In Section 7, research conducted on controlled environments is examined. We further divide the literature works into the targeted environment: office, campus, car and non-restricted daily life conditions and discuss prominent features and techniques for each environment in Section 7. Stress alleviation methods in the literature are briefly discussed in Section 8. We then conclude with the research challenges for the stress detection studies.

2. What is stress?

Stress is a widely used term but it is not easy to agree on the definition because it is a subjective and hard to define phenomenon [12]. However, if scientists cannot define stress, how could they quantify it? Merriam Webster dictionary defines stress as a physical, chemical, or emotional factor that causes bodily or mental tension and may be a factor in disease causation. Stress can be non-formally defined as the body's way of reacting to any demanding or hazardous situation [13].

The stress reaction is initiated by the brain as a response to sensory inputs from the eye, nose or ear. When the body senses a threat, it could be real or imagined, defensive mechanisms of the body initiates a rapid, automatic process called as the "fight-or-flight" reaction or the stress response to protect itself. Brain immediately sends a distress signal to the hypothalamus. Hypothalamus is analogous to the command center of the brain [14]. Hypothalamus controls involuntary body functions via the autonomic nervous system (ANS). ANS constitutes from two elements which are the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). SNS can be considered analogous to a gas pedal in a car[14]. In fight or flight situations, after getting the distress signal, hypothalamus activates SNS. SNS releases stress hormones such as epinephrine and cortisol, which arouse the body to act in emergency situations. Heart rate increases, muscles tighten, blood pressure increases, inhalation frequency rises when the SNS is activated. Volumes of the airways in the lungs increase [14]. Oxygen in the brain increases and this causes senses to become sharper [13]. Blood sugar is also increased which makes an individual more energetic. Abovementioned changes improve the strength and stamina, shortens the reaction time, and improve the focus. If this mechanism works as desired, it helps an individual to stay focused and alert. When the danger passes, PNS, which is analogous to the brake in the car, diminishes the stress reaction.

The abovementioned stress reaction is the physiological response of an individual to a stressful event and can be named as physiological stress. Another type of stress occurs due to mental appraisal and interpretation of the stressful situations by an individual and can be called as perceived stress [15]. Perceived stress can be measured with periodical self-reports collected from the individuals [15]. In other words, it is the perception of the stress from an individual's point of view. Theoretically, researchers expect that these two stress levels have a coherency between them. However, perception could be subjective, it can change from person to person. Liapis et al. [16] give examples regarding the subjectivity of the self-reports. They demonstrated that although physiological data is similar for both genders in their laboratory experiments, women tend to express stress level more than men on self-reported data. Another problem of perceived stress is that since the self-reports are collected at certain periods, they can miss the stress episodes in some cases. Stress has a negative effect on memory and people have a tendency to forget stressful events [17] especially if the survey collection period is long. These two problems may result in discrepancies between the measured physiological stress levels and perceived stress levels. Collection of perceived stress data is also more difficult than physiological data collection. Although for the collection of the physiological data, an unobtrusive wearable device can be sufficient for the perceived stress, participants should fill some surveys periodically which could create a burden on the individuals.

Stress helps the individual to survive in dangerous cases. Up to a certain level, stress can be also helpful in demanding situations such as presentation at work, exam in school. As abovementioned, stress is a mechanism of the body that helps and saves us in critical cases. However, after it exceeds a certain level, stress is not beneficial anymore, on the contrary, it starts damaging the health, emotional state, productivity and life quality of an individual. The reason is that our nervous system cannot differentiate between emotional and physical dangers [13]. In emotional demanding situations such as an argument with a friend or a deadline, the nervous system reacts in the same way as a daily life-or-death situation. If this activation becomes frequent and an individual stresses out more, body would be stressed most of the time, and this can result in serious health problems.

3. Stress impact on society and economy

Daily life stress has become a significant issue for the modern society. Offices among other places contribute to the high stress most [11]. The mismatch between job demands and abilities, time pressure and high workloads are general reasons for the office stress. Family-related issues, illnesses and chronic injuries and emotional problems can be listed as off-the-workplace stress causes. Stress is the second most severe work-related health issue in Europe [18], after musculoskeletal illnesses which can be caused by stress in some cases [19]. The American Institute of Stress reveals that the US spends 300 billion USD per year to diseases caused by stress [5]. In 2013, work-related stress cost 25 billion euros to the EU businesses [20]. Recent public survey [21] revealed that 51% of European workers are exposed to stress at workplaces. It is anticipated that 50–60% of all lost working days in the European business sector caused by work-related stress and psychosocial risks [11].

There are two types of stress: acute and chronic [1]. Acute stress is more common and the majority of people have experienced this kind of stress. American Psychological Association stated that demands and pressure from recent past and near future cause acute stress [22]. The potential triggers for acute stress can be listed as athletic challenges,

test taking, or anxiety when meeting new people. The causes of chronic stress can be counted as long-standing pressures and demands as a result of socioeconomic conditions, difficulties in interpersonal relationships, or an unsatisfying career [22]. The consequences of chronic stress can be destructive if left unmanaged [23]. Since the short-term symptoms of acute stress are more observable, researchers have investigated acute stress more than chronic stress. Furthermore, since the duration of acute stress is shorter, subjects are less affected when they are induced in the experiments when compared to chronic stress.

As abovementioned, stress has noticeable effects on human health. In acute stress, possible symptoms can be listed as emotional distress, muscular ache and tension, digestive tract issues, and overarousal [24]. Overarousal related problems are more prominent which can lead to heart attacks, arrhythmias, and possible sudden death in those with preexisting heart conditions [25]. Minor effects on physical conditions may include headaches, back pain, heartburn, stomach ache, elevated blood pressure, and rapid heartbeat [24]. Chronic stress has similar effects with acute stress. However, it causes more damage to physical conditions. It is a significant risk factor for hypertension and coronary disease [25,26], irritable bowel syndrome, gastroesophageal reflux disease [27], generalized anxiety disorder, and depression [28]. These health issues also have a prominent effect on the economy such as absenteeism, staff turnover [29] and tardiness. These problems result in a decrease in productivity. “Presenteeism” problem, which can be defined as employees are present at their workplace, but they do not work at full capacity has also been emerged due to stress [11]. The yearly cost of absenteeism and presenteeism has been estimated at 272 billion Euros and the cost of productivity loss has been 242 billion Euros every year [20] in the whole of Europe. The long-term consequences of stress triggered a need for avoiding it when the symptoms first emerge. To prevent more damages, it should be detected in early stages.

4. Stress induction tests

In order to investigate stress in laboratory environments, subjects are asked to perform certain tasks in order to induce stress. In this section, the most prominent tests applied in the laboratory conditions will be briefly described.

4.1. Trier Social Stress Test (TSST)

The most common stress induction test is the TSST. The procedure of TSST is as follows. The total period of the test is fifteen minutes [30]. It is divided into three phases. During the test, saliva and blood are collected and the heart rate data is recorded. The test takes place in a room with a video camera, microphone and three judges. Judges are prepared and trained for controlling their expressions as neutral. In the first phase, the subject is asked to prepare a presentation that has five-minute duration. The subject has a paper and pen to prepare the presentation. After the first phase, the paper is suddenly pulled back from the participant. In most versions of TSST, the subject is told that the presentation is for a job interview. In the second phase, the user is asked to present the prepared material. Judges listen to the presentation without any intervention. If the participant finishes the presentation before five minutes, they ask him or her to continue. The last phase is a mental arithmetic task in front of the judges. The subject is asked to count backwards from 1022 in steps of 13 [30]. If the participant makes a mistake, he/she is asked to begin from the start. After the last five-minute mental arithmetic task, recovery period starts. The collection of samples and signals continues until the end of the recovery period. At the end of the test, the user is briefed about the aim of the test.

4.2. Stroop color-word inference test

The Stroop Color-Word Test (SCWT; Stroop, 1935) is one of the oldest and most widely applied stress induction test [32]. By changing the number of subtasks, type and number of stimuli, times for the task or scoring procedures, different variations of the test were created [33]. In the standard version of the test, a subject is asked to read the name of the color words. This subtask is called “word reading”. The second subtask is naming the color of ink. The last subtask is to name the color of the word that is come up at the screen. An example of SCWT test is shown in Fig. 1.



Fig. 1. SCWT Test Example [31].

4.3. Montreal Imaging Stress Task

The Montreal Imaging Stress Task (MIST) is a variation of Trier Test [34]. It includes computerized mental arithmetic challenges and social evaluative threat elements with the help of a computer program. It is applied by an investigator. In order to be able to differentiate the effects of stress and mental arithmetic, it has three phases (rest, control and experimental). In the original test version the subject is observed via functional magnetic resonance imaging (fMRI) or positron emission tomography (PET) in these phases. In the resting phase, the subject looks at the computer display. In this phase, subjects are not asked to perform any tasks. In the control phase, a series of mental arithmetic tasks are shown to the participant and participant answer the tasks by using the provided computer program. In the last phase which is the experimental phase, the difficulty of the tasks is increased and the time limit of the tasks is decreased. The aim is to increase the task difficulty over the participant’s mental abilities. To increase the stress of the participant more, the average performance of participants and estimated completion time is displayed to the subject. After the end of each task, the performance evaluation is displayed to further increase the stress of the participant.

4.4. Cold pressor test

The cold pressor test is another cardiovascular test applied by putting the participant’s hand into an ice water container for a minute [35]. It can be also used to induce stress. Blood pressure and heart rate are monitored. The procedure of the test is as follows: The subject of the test is asked to put his/her hand in the cold pressor as long as they can. They are told to express if they feel any pain. If the pain becomes unbearable, they can remove their hands. This helps researchers to measure the threshold (first feeling pain) and tolerance (total time minus threshold). This version of the cold pressor task is the most widely applied one.

4.5. Sing-a-Song Stress Test (SSST)

Brouwer et al. [36] developed a new method for inducing stress which is named SSST. The subjects are shown neutral messages at the screen for one minute. After this neutral phase, a final message that is asking the subjects to sing aloud is shown. The participants must stand still during the singing. They remarked that SSST is a quick, easy,

controlled and potent way to induce mental stress [36]. They measured the ECG and EDA signals during both the neutral and singing phases and they demonstrated that two phases are considerably different and found a correlation between the stress level and singing.

4.6. International Affective Picture System (IAPS) test

The IAPS is an image database created for providing a standardized set of images for emotion and attention research [37]. It has been commonly applied in psychological studies [38]. The IAPS was created by the National Institute of Mental Health Center for Emotion and Attention at the University of Florida [39]. It consists of pictures varying from simple household objects to extreme pictures which cause arousal on individuals (mutilated corpses, erotic and violent scenes). Stress induction by showing a series of pictures from the IAPS is another test that is widely applied among the literature.

5. Stress signals

We can measure and observe stress symptoms in numerous ways. The Sympathetic Nervous System (SNS) ignites the stress reaction [40], resulting in psychological, physiological and behavioral symptoms [41]. The psychological way of measuring stress can be self-report questionnaires or being interviewed by a psychologist [11]. Therefore, automatic stress detection topics do not include this class.

The second way to detect stress is by evaluating physiological signals (see Fig. 2 for different types of them). They include information

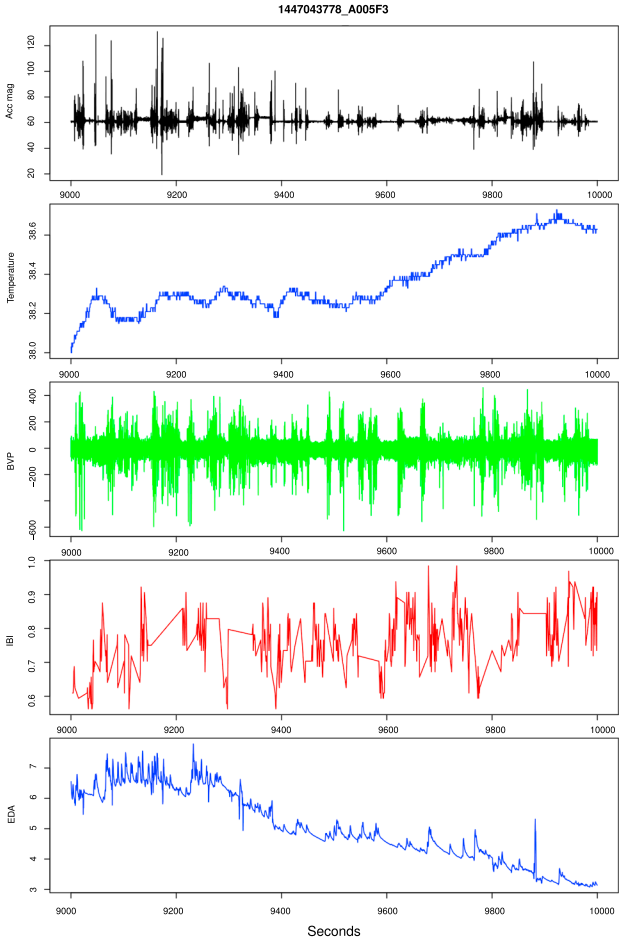


Fig. 2. We collected different types of physiological signals from Empatica E4 (From Top to Bottom: Accelerometer, Skin Temperature, Blood Volume Pressure, InterBeat Intervals and Electrodermal Activity Data).

related to intensity and quality of the affect and experience of the subject [42]. Signals of interest include hormone levels, Electro-Cardiogram (ECG), Electroencephalogram (EEG), Electro-Dermal Activity (EDA), Blood Pressure (BP), Skin Temperature (ST), Electromyogram (EMG), Respiration, Blood Volume Pulse (BVP), Pupil Diameter (PD), Eye Gaze and Blinking, Thermal Imaging (TI) and functional Magnetic Resonance Imaging (fMRI). The other two methods are investigating behavioral data and context information which are not investigated thoroughly in the literature [43].

Another way of recognizing stress is from the behavioral changes. Stress affects in behaviors of individuals. Without invasive methods and a need for extra pieces of equipment, we can measure the behavioral changes. Behavioral responses comprise of keystroke and mouse dynamics, posture, facial expressions, speech, mobile phone usage, walking pattern and text linguistics. The last property we can recognize stress from is the contextual information which consists of calendar events and location. In this section, we will discuss the most important signals for stress detection, widely used types of equipment for each signal and the most distinctive extracted features.

5.1. Physiological signals

5.1.1. Heart activity

One of the most prominent signals for discriminating stress is the heart activity because ANS influences the heart rate directly. The electrocardiogram (ECG) is employed to measure the electrical activity produced by the heart via electrodes placed on the body typically to the left arm, right arm and the left leg. A typical heart beat has four fundamental elements which are baseline, P wave, QRS complex and T wave [8]. The most distinctive R peak is employed for feature extraction. Heart activity can be modeled with heart rate (HR), RR intervals (IBI) and heart rate variability (HRV). Heart rate variability is the oscillation of the time between consecutive beats. IBI interval can be defined as the time between two consecutive R peaks. All of these can be inferred from R peaks. The wearable devices that are used commonly for ECG measurement are Biopacs MP150, MP35 and Shimmer Sensing 3 [8]. Features can be divided into three classes: time domain, frequency domain and non-linear features. Time domain features are the mean of RR intervals, the standard deviation of RR intervals, the root mean square of RR intervals, the percentage of the number of successive RR intervals varying more than 50 ms. Frequency domain features can be counted as Low-Frequency component, High-Frequency Component, LF/HF ratio. Most common non-linear features are entropy, complexity, Poincare Plots, recurrence and fluctuation slopes [11].

5.1.2. Brain activity

Brain activities are also affected from emotional changes and stress. The electroencephalogram (EEG) is used to measure the brain activity by placing a series of electrodes onto the scalp of the subject. BioSemi ActiveTwo, ABM B Alert X10, Emotiv EPOC+ are widely used gold standard types of equipment for measuring EEG. EEG signal is composed of four frequency bands: Alpha (8–13 Hz), Beta (13–30 Hz), Delta (0.1–4 Hz) and Theta (4–8 Hz) [11]. Alpha activities are a sign of a calm and balanced state of mind and decrease in stressful states. Beta activity correlates with emotional and cognitive processes and increases with stress. Mean amplitudes of the EEG signal, mean amplitudes of Event-Related Potentials (ERP), alpha, beta, theta frequency bands, mean power ratios and fractal dimension features are used to detect stress.

5.1.3. Muscle activity

The stress-related neural activity also affects muscles. Muscle action potentials are used to detect stress [8]. Electromyogram (EMG) measures muscle action potentials by placing electrodes on selected muscles. Facial and Trapezius muscles are regions of interests for measuring muscle activity. The mean, median, standard deviation, RMS, peak loads and gaps per minute are the features that are used commonly [11].

5.1.4. Electrodermal Activity (EDA)

EDA, also known as Galvanic Skin Response (GSR) is the change of electrical properties of skin. Under emotional arousal and stress, body sweats and skin conductance increases. EDA can be computed by applying a small current and measure the resistance of skin between two placed electrodes. The EDA signal is composed of two components. The first one is the Skin Conductance Level (SCL). It represents slowly changing part over a long term (Tonic). The second part is the Skin Conductance Responses (SCR) which represents the faster and event-related part of the EDA (Phasic). To measure a slowly moving baseline and extract statistics related features such as mean, standard deviation and percentiles researchers are using the Tonic (SCL) part since this part is not contaminated with peaks which will affect the baseline calculations. For arousal event detection, the Phasic (SCR) part is employed. EDA is one of the best discriminative signals along with the heart rate signal. ProComp Infiniti, Biopac MP150, Shimmer 3 GSR+ and Empatica E4 wristband are the instruments that are used widely to measure the EDA [8]. The mean amplitude, standard deviation, minimum and maximum values, RMS, the delay between applied stimuli and response, number of peaks, peak height, rising time, recovery time, the position of maximum and minimum features are tried in the literature to determine stress of a user [11].

5.1.5. Blood volume pulse

When HR and HRV change with stress stimuli, blood volume and blood pressure also change. Blood volume pulse (BVP) is the change in the blood volume for each interbeat interval. Photoplethysmography (PPG) is the low-cost optical technique to measure BVP. It uses the absorption of light by blood. After light is emitted from a light source, different amounts of blood in the volume will absorb different amount of light. In this manner, blood volume can be measured. UFI model 1020, Empatica E3 and E4 wristbands, Angel Sensor, Biopac Bionomadix PPGED-R are the pieces of equipment that are commonly employed to measure BVP by using PPG. Although BVP features can be used directly, in general, they are used to extract the heart rate variability or IBI features.

5.1.6. Skin temperature

Skin temperature can change due to various factors including stress. Research demonstrated that arousal can cause 0.1 or 0.2 Celsius temperature change [44]. The main reason for local temperature changes is that the blood flow is controlled by SNS. By controlling other factors, the effect of stress on the skin temperature can be measured. Mean, minimum, maximum and standard deviation of skin temperature features were used in the literature to determine stress.

5.2. Behavioral data

5.2.1. Speech

Stress causes changes in the human voice generation mechanism. Pitch, speaking rate, energy and spectral characteristics are affected by stressful events. Speech is preferred by many researchers because it is noninvasive and data collection is easy in the controlled quiet environments. Pitch (mean, standard deviation, range), higher frequency bands ratio, speaking rate, voice intensity, smoothed energy, voiced-unvoiced speech ratio, Mel Frequency Cepstral Coefficients (MFCC) are the features that are used widely to detect stress levels [11].

5.2.2. Facial expressions

Stress and emotional states have a correlation with facial expression. It has been demonstrated that facial expressions reflect emotions more than self-reports [45]. Facial EMG and image recognition from cameras were used to detect facial expressions in stressful situations. Mean smile intensity, eyebrow activity and mouth activity are the main facial features for detecting stress [11].

5.2.3. Keystroke and mouse dynamics

Another important behavioral data is keystroke and mouse usage dynamics. Every individual has a different writing and mouse usage speed and style. Especially the speed of writing and mouse usage depends much on individuals [46]. These two features define the keystroke and mouse dynamics of a user. When an individual is stressed, the muscles contract more than usual and it affects keystroke and mouse dynamics of an individual. Stress can be detected by relying on this change. Keystroke and mouse dynamics is a noninvasive technique and does not require additional equipment other than the mouse and the keyboard. The most important and measurable features that distinguish stress from keystroke dynamics are dwell and flight times, pause rate, the frequency of using specific keys such as backspace and space-bar, duration of digraphs and trigraphs and key pressure [11]. Important mouse dynamics features can be listed as the acceleration, average speed of the movement, frequency of the movement, stillness, frequency of the clicks.

5.2.4. Body gestures and movements

Individuals demonstrate behavior and gesture changes when feeling stressed. These changes include but are not limited to jaw clenching, arm movements, self-touching, finger rubbing. Posture change is another sign of stress. Stress in sitting position is investigated by the change in the center of pressure [47]. Subjects revealed more posture changes under stressful situations.

5.2.5. Mobile phone usage

Stress also affects how the individual uses smartphones. Since collecting information is easy and non-invasive, smartphone usage behavior changes are investigated in the literature. Call logs, SMS logs, app usage, types of apps, battery usage, the screen on–off frequency, internet browsing and Bluetooth proximity were used to detect stress [11].

5.3. Questionnaires and surveys

Psychological stress evaluations can be collected by asking subjects to fill questionnaires and by interviewing. There are two ways to collect data from subjects in daily life (or long laboratory experiments) which are instant reporting and day reporting (or cumulative reports). People have a tendency to forget emotional peaks in 24 h [48]. For this reason, asking questions to subjects at the end of the day can cause incorrect measurements. The alternative is to ask subject to report stressful events instantly. The problem in this manner is that people can forget to report events. In the literature, a combination of these methods is used to increase the accuracy of reports. Perceived Stress Scale (PSS), Stress Self-Rating Scale (SSRS), NASA-TLX, Self Assessment Manikin and Positive and Negative Affect Schedule (PANAS) are the questionnaires and interviews that are commonly used in the laboratory and daily life stress experiments.

6. Data collection challenges

In the laboratory environments, data collection procedures are less prone to error when compared to daily life data collection. In the daily life, errors from incorrect placement or detached equipment can occur. Corrupted physiological signals due to body movements are very common. Multi-sensor, multi-device measurements, invasive devices and storage and processing of the massive amount of data can be counted as the additional challenges of data collection.

6.1. Movement and improper placement problems

High-quality data must be accurate, complete, relevant, timely, sufficiently detailed, appropriately represented, and must retain sufficient contextual information to support the decision making [49].

Wearable physiological measurement devices mentioned above satisfy most of the listed conditions and provide high-quality data. The sampling frequency should be sufficient to represent the signal well. Proper placement of the sensors should be done to avoid ambiguities and obey standards for the correct measurement of physiological data. Signals are affected by the device noise, random noise as well as by loose device-skin connections and body movements [11]. Noise in the signal can be filtered by Kalman filters, Butterworth low-pass filters, Median filters, Wiener filters, Wavelet Decomposition, etc. [11]. Artifact removal can be done via least mean squares, regression analysis, independent component (ICA) and principal component analysis (PCA) [11].

6.2. Data fusion from variety of sensors

In most of the experiments, data is retrieved from different sensors and devices. Integration of the data has challenges. Determination of the point of data integration, i.e., before the final decision or during processing is the first challenge. Synchronization of the data must be achieved via timestamps for the calculations to be accurate. The last challenge is the missing data from one source or more sources. Some precautions must be taken such as using probabilistic data fusion algorithms, imputation or removal of the missing time intervals.

6.3. Big data problem

Continuous data collection accumulates a massive amount of data to be processed and stored. Onboard signal processing algorithms could be a solution to the big-data problem. Nevertheless, this increases the power consumption, decreases the battery life of the devices, reduces storage requirements but increases the algorithm complexity [11]. Researchers need to cope with this trade-off according to their experimental conditions.

6.4. Selection of non-invasive devices

Instruments to measure stress should be unobtrusive and non-invasive to collect data accurately. Invasive devices can create additional stress on subjects when they are worn. Recent technologies have provided us non-invasive and completely transparent devices to the user for monitoring stress. Smart wearable systems and smartphones are commonly used to collect significant amounts of data non-invasively and sometimes without the user being aware of it.

6.5. Battery life

Charging is also a significant issue in real-life data collection. When the life of the battery becomes three or four hours (as in the case of Samsung Galaxy Gear S1, S2, S3 when all of the sensors are active), devices must be charged several times for collecting one day long continuous data. Besides being inconvenient, the need of recharging may lead to data gaps between the times when devices are collected for recharging and handed out to the participants with the full battery again. The battery issue directs researchers towards new challenges such as designing batteries with longer lives on wearable devices, increasing the life of the battery by reducing the power consumption (i.e., disabling some sensors, decreasing the duty cycle of sensors, cutting down the brightness levels).

6.6. Ground truth collection

One of the most important differences between the research that is conducted in the laboratory and in daily life is the need for collecting the conditions of the participants. In the laboratory experiments, the condition of a subject is always known because the experiment steps are designed with the timestamps beforehand. However, in the daily life data collection, the stress condition of a subject is unknown a priori and the only way to learn this ground truth is to ask the participants. For this purpose, some known surveys (Perceived Stress Scale (PSS), Stress Self-Rating Scale (SSRS), NASA-TLX, STAI, Self Assessment Manikin and Positive and Negative Affect Schedule (PANAS) questionnaires) are employed during a day with the determined time periods. The surveys must be distributed to all of the participants and collected from all of the participants for each session of the daily life experiment periodically. Another alternative might be using a mobile survey app developed for this purpose. The ground truth collection from the participants imposes some new challenges on researchers. The scores are subjective and change from person to person. They may not represent the true stress status of a person due to the difference between perceived and physiological stress as mentioned in Section 2.

7. Stress detection: Literature review

In this section, we provide a literature review of the stress detection studies. Firstly, the controlled laboratory environment where the research has first started is investigated. This section is divided into subsections by taking the used physiological signal into account: heart activity, electrodermal activity, brain activity, speech data, camera-based studies and multi-modal measurement research. The works that use stress alleviation methods after detecting stress are presented at the end of the section.

We classify outside the laboratory environments as restricted, semi-restricted and unrestricted. Automobile and office are restricted environments since the movement is limited and environments can be controlled with sensors and cameras. University campuses are semi-restricted environments. Some parts of campus environments are more constrained (i.e. classrooms, offices) and some parts are less constrained. This section is examined as a transition between restricted and unrestricted environments. Lastly, we present the literature in unrestricted daily life. We provide an insight subsection for each environment which includes the most successful machine learning (ML) algorithms, discriminative physiological signals and features. We also highlight the outstanding studies for each environment.

7.1. Controlled laboratory environment: early works

Stress detection research in the laboratory environments are the preliminary works that give researchers ideas about which sensors, features and machine learning algorithms to use in daily life studies. In most of the laboratory studies, ground truth collection methods are not required because the cognitive load and relaxation activity of the participants are known a priori. We will examine these works by dividing them according to the used sensor type.

7.1.1. Heart activity

Heart rate variability is one of the most distinctive signals for stress detection. A heart activity measuring device for the laboratory environments is shown in Fig. 3. Time domain (mean RR, RR standard deviation), frequency domain (LF/HF, normalized LF HF difference) and nonlinear features were used in the literature. See Table 1 for the summary of heart activity stress detection literature. Boonnithi et al. [53] investigated HRV features from both time and frequency domains. They used each feature with a minimum distance classifier and presented the classification accuracy for differentiating stress and relax state. They found out that the mean RR, the mean heart rate, normalized low frequency, the difference of normalized low and high frequency and the sympathovagal balance index (SVI) were the most distinctive features for stress detection. Costin et al. [50] added morphological variability features to the abovementioned features. Morphological Variability (MV) is a complementary technique to enhance results from HRV features by using changes in beat morphology data [50]. They achieved a 10% increase by using MV from 80% to 90% classification accuracy in 3-class (low, medium, high) classification.

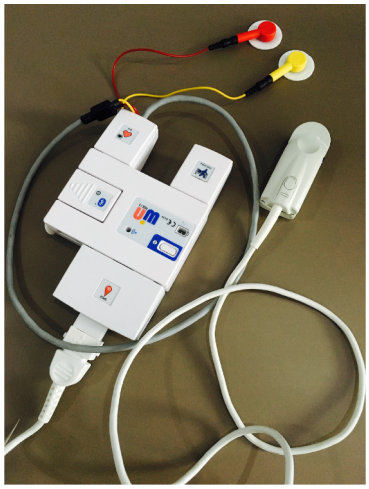


Fig. 3. An instrument for measuring position, HR (2-lead ECG) and SpO2 in laboratory environments.

Castaldo et al. [51] experimented ultra short HRV (2 min) features and their effect on the stress detection accuracy instead of short HRV features (5 min). They designed SCWT, daily life exam and video game tests at the laboratory. By applying SVM (Support Vector Machine) over HRV features, they achieved 70% accuracy with the binary classification (stress vs. relax) during laboratory conditions. These results are lower than the literature results. They inferred that this could be due to stress induction tests or ultra short-term HRV features.

Stress recognition by employing HR features was also used to improve an emotion recognition system. Tivatansakul et al. [52] developed a system that recognized negative emotions from facial expressions and provided hints on how to relax. However, they stated that the system has a confusion problem when differentiating positive and negative emotions and the binary classification accuracy was 62.5%. To improve the system, they added a stress detection scheme using the ECG signal and HRV features. Participants were exposed to some negative images for four minutes and after that, features were extracted and an SVM classifier was applied. By combining the ECG based negative emotion and stress detection scheme and emotion recognition scheme which uses facial cues and evaluating stress and negative emotions together, they achieved 83% negative emotion classification accuracy which is approximately a 20% increase.

7.1.2. Electrodermal activity

EDA signal is one of the most used signals for stress detection along with the heart rate signal. An EDA measurement instrument is demonstrated in Fig. 4. One pioneer research using EDA was carried out by Bakker et al. [56]. They first determined possible problems of wearable EDA devices as subjects are touching devices, subjects wearing devices not tight enough and unrelated EDA responses from physical activities and an increase in temperature. They used manually added stress labels to EDA data as a reference. By employing ADWIN fit method [57], they developed a successful stress detection scheme. As future works, they plan to add calendar, e-mail and social media data to classify stress better.

Another EDA-based research was carried on by Bornoiu and Grigore [58]. They focused on EDA feature extraction and filtering part which they claimed missing on the literature. Stress was induced by using the MIST test. They first applied a moving average filter to eliminate the high-frequency components. They further applied Fourier transform and low pass filter on the frequency domain to get SCR (Skin Conductance Response). They converted the signal to the time domain by employing the Inverse Fourier Transform. By extracting SCR from the EDA signal, they got SCL (Skin Conductance Level). Another important observation they made from a 31 subject experiment was finding out that the response time to the stressors has a Gaussian distribution with a mean 3.32 s and standard deviation 1.4 s.

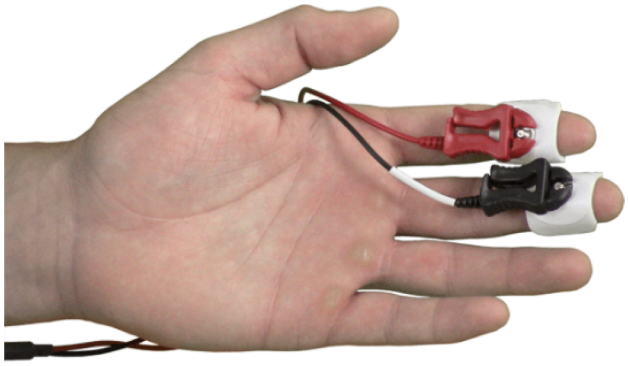


Fig. 4. EDA measurement instrument for laboratory environment [59].

EDA signals were used for discriminating stress from the cognitive load in [54]. As the control condition, Setz et al. employed a mental task which was used to measure only the cognitive load. They further added a time pressure and social threat which is MIST test to measure stress plus the cognitive load. They collected EDA, ECG, breathing and movement data from the subjects. Furthermore, for this application to be more suitable for future everyday life application, they used EDA for sensor setup. A mobile device the so-called Emotion Board was used to collect EDA data. Mean EDA peak height, peaks per minute, EDA slope, mean, maximum and minimum were used as features. Linear Discriminant Analysis (LDA), SVM-RBF and Nearest Class Center (NCC) were the data mining algorithms employed for the classification. They achieved 82.8% accuracy while discriminating stress from the cognitive load which is a more difficult task when compared with classifying stressful and stress-free situations.

The effect of gender differences on responses to stressors was also investigated in the literature. Liapis et al. [16] measure physiological signals and collected self-reported data on both genders. They collected EDA data on different HCI tasks. They created five HCI scenarios: missing file, hardware failure, slow network speed, popup advertisement and information seeking namely. LDA was the most successful classifier for both genders (94.8% on males, 98.85% on females). They concluded that although physiological data is similar for both genders, women tend to express stress level more than men on self-reported data.

Human activities can affect the success of automatic stress detection algorithms. Tang et al. [60] developed an activity recognition algorithm to improve the stress detection scheme which uses a GSR sensor. By utilizing the accelerometer sensor, they classified sitting, standing and walking activities with 94% accuracy. 5 healthy male subjects which were a quite limited number of participants when compared to the literature. MIST method was used as the stress induction method. They

Table 1
Stress recognition using heart activity in laboratory environments.

Article	Stress signal	Stress test	Method	# of Classes	Accuracy %
[50] (2012)	HRV	–	Minimum Distance Classifier	3	90
[51] (2018)	HRV	SCWT	SVM	2	70
[52] (2015)	HRV	IAPS	SVM	2	83 (Negative -Positive Emotion by using Stress Detection)

Table 2
Stress recognition using electrodermal activity in laboratory environments.

Article	Stress signal	Stress test	Method	# of Classes	Accuracy %
[54] (2010)	EDA	MIST	LDA – SVM-RBF – NCC	2	82.8
[55] (2015)	EDA – Accelerometer – Bluetooth	–	Logistic Regression	2	91
[16] (2015)	EDA	–	LDA	2	98.8

showed that with a little increase in false positives, the accuracy of the stress detection algorithm was improved.

Zubair et al. [55] designed a smart band that includes EDA sensor, accelerometer and Bluetooth to detect stress of an individual. They used EDA and accelerometer together to increase accuracy. The experiment was in a controlled environment with twelve subjects. By exploiting accelerometer data, they classified human activity and utilized the activity to improve the stress detection system performance. They discriminate between stress and non-stress by using a logistic regression classifier. They achieved 91% classification accuracy. Table 2 summarizes stress recognition works using electrodermal activity in the laboratory environments.

7.1.3. Combination of electrodermal and heart activity

One of the most commonly used signal combinations to detect stress is the HR and EDA combination (see Table 3). Sierra et al. first proposed a stress detection scheme for real-time applications [67]. They asked subjects to hyperventilate first then prepare a talk. They recorded HR and EDA signals during these tasks. KNN (k Nearest Neighbor) and Fisher discriminant analysis were applied to detect stress. They achieved approximately 95% classification accuracy when differentiating between stress and relax states on average. They stated that 5–10 s intervals are suitable for real-time applications. In [66], they extended their work on [67] by employing fuzzy logic. They defined the calm state, stimulating task, threatening task and baseline post-stress stages and recorded signals. They applied manual (proposed by Mamdani) and automatic (proposed by Sugeno) implementations of fuzzy decision algorithm to detect stress. In the manual implementation, they set the system parameters apriori and in the automatic implementation, system adjusted its parameters by learning. They achieved 99% classification accuracy with two class classification (stress vs. relax). Automatic implementation outperformed the manual implementation. They further investigated the effect of the template period to stress detection success. They stated that longer periods would increase the accuracy of a system.

Sandulescu et al. [64] developed a stress detection scheme using BioNomomadix module from Biopac. They collected the combination of PPG and EDA signals. SVM was used as a classifier. They induced stress by using the TSST (See Section 4.1). With 5 participants, they achieved 80% classifying accuracy when discriminating between stressful and non-stressful situations.

Ollander et al. [68] compared wearable sensors with the stationary sensors measuring PPG and EDA signals which have the same functionality. TSST was applied to induce stress. The authors first compared Empatica E4 Inter-Beat Interval (IBI) measurement with ECG. The missed heartbeats from the Empatica E4 increased during stressful tasks. They extracted time and frequency domain features from the physiological signals obtained from the Empatica E4 smart band. The mean and standard deviation heart rates extracted from E4 are very close to the real values. As far as stress detection is concerned, Empatica E4 and medical grade ECGs have very similar classification accuracies. The second comparison was between the E4 wrist skin conductivity and finger skin conductivity from the laboratory instrument. They demonstrated that the stress detection power of E4 is higher than the laboratory sensor. Weaknesses of this article can be counted as the limited number of participants (7 participants), conducting the experiment in a laboratory environment instead of daily life.

Martinez et al. also worked with HRV and EDA signal combinations using Biopac-Mp36 devices [63]. 166 participants were selected from the university. Puzzles were used to induce stress. During the experiment, signals, questionnaires and expert counseling from psychologists were collected. A classification algorithm based on F- state machines was implemented. Finite state machines are defined as the mathematical model of computation [69]. 0.984, 0.970 and 0.943 f-measure scores for high, medium and low-stress levels were achieved.

Mozos et al. [61] combined physiological and sociometric sensors to detect stress. As physiological sensors, they used EDA and PPG. Sociometric sensors are microphone and accelerometer which are included in the sociometric badge used to detect the social activity. They measured the baseline and used TSST for inducing stress. Subjects are asked to prepare a presentation and to give a public speech. They used personalized classifiers for individuals. kNN, SVM and AdaBoost were applied as classifiers. Adaboost also measured the strength of features when discriminating between stress and neutral situations. The amplitude and frequency of voice, EDA, body movement activity and posture activity were found to be the most discriminative features. Adaboost gave the highest accuracy results in the personalized classification (94%). In the person independent classification, they obtained 67% classification accuracy. They plan to extend this research into daily life scenarios.

7.1.4. Speech data

Speech data can be also used to detect stress levels of the individuals. Kurniawan et al. [40] developed a combined system using both Galvanic Skin Response (GSR) signals and speech of an individual. They created a ten people benchmark dataset recorded in a laboratory environment. Stroop Color-Word Interference Test, TMST (Trier Mental Stress Test) and TSST (Trier Social Challenge Test) tests are used for

Table 3
Stress recognition using heart activity and electrodermal activity in laboratory environments.

Article	Stress signal	Stress test	Method	# of Classes	Accuracy %
[61] (2017)	EDA – PPG – Speech – Accelerometer	TSST	Adaboost	2	94
[62] (2017)	ECG – GSR – respiration – Blood Pressure – Blood Oximeter	Ice test and IAPS	SVM – kNN	2	95.8
[63] (2017)	HRV – EDA	Puzzles	F-state Machine	3	0.984 (f-score)
[64] (2015)	EDA – PPG	TSST	SVM	2	80
[65] (2011)	ECG- EMG – EDA	–	Bayes, kNN, Fisher Least Square	2	80
[66] (2011) (Extension of [67])	EDA – PPG	–	Fuzzy Logic	2	99
[67] (2010)	EDA – PPG	–	kNN – FDA	2	95

benchmark creation. From the speech signals, pitch and MFCC features are extracted. Furthermore, sudden changes are recorded from EDA sensors. K-means, GMM (Gaussian Mixture Model), SVM and decision tree classifiers are used. Both sets of features were used separately and in a combined manner. The SVM classifier performs the best for each stress test. Speech features alone gave better classification results than GSR; however, they were more person dependent. The classification accuracy does not increase meaningfully when two features were combined.

Soury et al. also used audio features for stress detection [70]. TSST was employed in the form of public speaking task. Low-level descriptors (LLD) and window-wise Functional Descriptors (FD) were extracted from the sliding windows. Features were computed for the window itself and for neighbor windows (before and after). Features were pitch related (F0, silence, vocalization), energy-related, voice quality features (jitter and shimmer), speech and silence ratios (HNR), spectral (LPC and MFCC). An SVM classifier was applied. Larger windows (4–5 s) were selected for achieving higher classification rates. Expert Likert scale ratings (1–5) were used as a reference. They achieved 72% unweighted average recall rate in the stress detection task. For future works, they plan to add different speaker profile and task types (speech, negative questions, positive questions).

7.1.5. Brain activity

Brain activity is also a distinctive signal for stress detection. Kalas et al. [88] detected stress of the subject by using EEG signals (see Fig. 5 for an example EEG measurement instrument) and if the subject is stressed, they intervened to reduce the stress. There were ten subjects and eight channels of EEG were utilized. The classification algorithm was k-means. They set a stress detection threshold and compared the features with it to determine whether the subject is stressed or not. If a subject is stressful, they gave a low complexity puzzle to reduce the stress level. The novel part is the introduction of stress reduction manners.

Vanitha et al. [72] proposed another EEG based stress detection scheme for students. They experimented with six students and Emotiv Epoch headsets. Self-reports were collected by NASA-TLX questionnaire. Stress was induced by asking mathematical questions with different difficulty levels in a limited time. For feature extraction, Hilbert Huang Transform (HHT) was applied. SVM, LDA, QDA (Quadratic Discriminant Analysis) and kNN classifiers were used and they achieved 89% classification accuracy for 4 classes of stress with the SVM classifier. The weaknesses can be listed as the controlled lab environment and usage of an obtrusive technique EEG.

7.1.6. Camera-based stress detection schemes

Video and depth processing is also used for detecting stress (see Table 5). Aigrain et al. [73] asked subjects to answer the time-constrained arithmetic questions in front of a camera. Subjects filled the questionnaire after the task is completed. They extracted features from a Kinect video. Body activity, high activity periods, posture changes, detection of self-touching, facial features were extracted. SVM was applied as the classifier to discriminate stress from non-stress. They achieved 77% classification accuracy after person-specific normalization.

Aigrain et al. [74] improved their work by incorporating physiological measurements: cardiac features, GSR, EMG and respiratory features. They combined behavioral and physiological data. Another novelty was adding the external and psychology expert assessments in addition to the self-assessment reports. For the external observer assessment, they used crowdsourcing. People can add assessments by watching videos using the CrowdFlower platform. They showed that adding physiological features in addition to behavioral features increased classification accuracy to approximately 85% with an SVM classifier in 2-class classification. The best features for all assessment types were found. They added face touching and finger rubbing features. For future works, they plan to improve physiology and behavior coherence and develop a multi-label classification task.

Giannakakis et al. [75] developed a system for stress and anxiety detection from facial expressions. Eye motions, mouth activity, head-related motions and heart rate were measured from the camera. They employed camera based PPG from the facial cues to estimate heart activity. They designed a laboratory experiment composed of social exposure (foreign language speaking), emotion recall, stressful multimedia phases. There were twenty-three participants. With the above-mentioned features and kNN, SVM, Naive Bayes and AdaBoost classifiers they achieved 91% accuracy with three class classification, i.e., neutral, relaxed, stressed.

Cho et al. described their stress detection system in [80,81]. They utilized a low-cost thermal camera to detect breathing. Respiration variability spectrogram was used to extract features (see Fig. 6). SCWT and Mental Computation tests were applied to induce stress on the participants in a lab room. Feature scaling based on maximum and minimum scores was carried out to normalize subjective ratings. With a Convolutional Neural Network (CNN) classifier, they achieved 84.59% classification accuracy on binary classification and 56.52% accuracy on 3-class classification. The drawback of this work can be the difficulty of applying thermal imaging in daily life and unconstrained settings.

Pupil diameter marker was also used to detect stress in [89]. Torres-Salomao et al. asked subjects mental arithmetic questions in front of a camera and they observed the pupil diameter. They had to preprocess and filter out blinking and head movements to get more clear data. They stated that the Pupil Diameter Marker is a good indicator for detecting stress. However, it should be combined with other distinctive physiological measures such as heart activity or electrodermal activity to have higher classification accuracy. Mokhayeri et al. [76] also proposed a stress detection scheme by tracking the eye. They detected the



Fig. 5. EEG measurement instrument for laboratory environment (Emotiv EPOC+) [87].

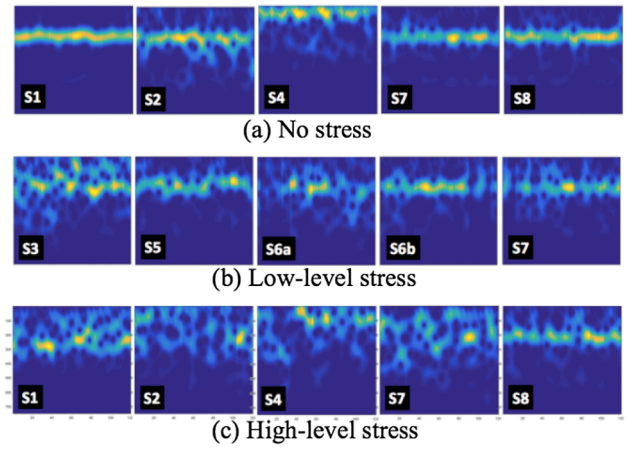


Fig. 6. Thermal camera spectrogram images of respiration from no stress, low stress and high stress respectively [80].

eye region via a Genetic Algorithm and to filter the noise, a fuzzy filter was applied. The Stroop color-word test was employed to induce stress. They used a fuzzy SVM classifier. They achieved approximately 70% accuracy in 2-class classification which is comparable with their PPG and ECG results.

Baltaci et al. [77] collected both pupil dilation and periorbital temperature data and extracted absolute and entropy-based features from this data. IAPS (International Affective Picture System) was applied to subjects for inducing stress. Approximately 90% classification accuracy in 2-class classification was achieved by using a decision tree classifier. They extend their work in [90] by adding entropy features for rising and falling of physiological signals in a time window and tested with AdaBoost with Random Forest Classifier (ABRF). They claimed that ABRF performs better than the decision tree classifier.

Human gaze behavior was another investigated feature to detect stress. Huang et al. [85] proposed a combination of human gaze and mouse click behavior scheme for this purpose. They aligned these two signals. Their stress inducing task was the time pressured arithmetic task. After the subject presses a show button, the arithmetic problem is displayed and gaze and click patterns are recorded. They extracted eight eye features which are calculated relative to the mouse clicking events. The Random forest algorithm was applied for a 2-class classification. They achieved 66% classification accuracy for individual specific models, 60% for the general model. They stated that although the relation between the individual gaze-click pattern and stress levels seems ambiguous, it could still be used for detecting stress.

Facial blood flow change is another characteristic parameter for stress detection. Thermal and visible cameras can be used for this purpose. Mohd et al. [83] developed a stress detection scheme by using these two types of cameras. They stated that facial thermal features (e.g. forehead) are difficult to obtain because of the low contrast. They focused on the nostril area by applying a Nostril Mask. The Graph Cut algorithm was applied to filter the data. They used both thermal and visual domains. The Scale Invariant Feature Transform was employed to extract the features. They achieved 88.6% classification accuracy in two-class classification.

Facial hyperspectral imaging (HSI) method was also used to detect stress [84]. HSI is an imaging technique making use of hundreds of narrow wavebands of the spectrum to create image cubes in both spatial and spectral dimensions. Chen et al. [84] obtained tissue oxygen saturation (StO2) data to detect stress. Thermal Imaging (TI) and HSI methods were compared. They demonstrated that they are consistent with each other and HSI is more robust than the facial TI because it is independent of perspiration and sudden environmental temperature changes. TSST test was applied as a stress induction method which includes mental arithmetic, public speaking and memory tests with 21 subjects. They further showed that StO2 levels in eye sockets and the forehead are discriminative features for stress detection. They achieved 76% classification accuracy with automatic threshold selection and 88% with manual threshold selection with 2-classes.

7.1.7. Multimodal measurements

Researchers have employed multimodal measurements to improve their stress detection accuracies. One activity-based research was carried out by Hong et al. [78]. They modeled stress for different physical activities instead of building a general model. They defined three activities: sitting, walking, cycling and they modeled physiological signals in the stressed situation during these activities. They used the NASA-TLX test as a subjective measure. They used Bioharness BT and SenseWear armband to collect physiological signals which are 3D acceleration, breathing, ECG, skin temperature, heat flux, galvanic skin response and near-body ambient temperature. Applied classifiers were Naive Bayes and Bayesian Networks. They achieved 82% classification accuracy based on an objective measurement for two classes.

In [79], authors extended their work on [78]. They made their work more suitable for daily life conditions. They induced stress in the same way by applying noise, cold water and verbal math. They stated that in everyday life tagging activities and training a supervised model would be difficult, open to activity classification errors and three activities are too limited. They trained an unsupervised model which can differentiate high activity and low activity. They build a model based on these two classes. They also clustered people who respond similarly to the stressors instead of building models for individuals as in [78]. They grouped activities by using k-means, Naive Bayes and Logistic Regression classifiers. They obtained 65% classification accuracy which is less compared to their previous work but more practical for daily life applications.

Akhonda et al. [82] measured stress on computer users. They collected EEG, ECG, EMG and EOG data. EEG data and psychological test data were collected at the beginning and the end of a session whereas ECG, EMG and EOG were collected continuously. They created three classes: stress due to eye strain, stress due to mental stress and resting state. A neural network was used as a classifier. For 12 subjects, they discriminate between three states with approximately 80% accuracy.

Keystroke dynamics (KSD) was another feature that was employed to detect stress. Since stress has an effect on many of the muscles around arms, neck and shoulder, typing patterns should be altered. Gunawardhane et al. [91] extracted three features related to keystroke dynamics to detect stress: durations between key pressers of specific digraphs, trigraphs and error rate of backspace and delete. As references, they used a questionnaire and HRV monitoring device. They showed that KSD is a good indicator of stress by applying the Wilcoxon Signed Data Rank test.

7.1.8. Stress detection and alleviation systems

In addition to detecting stress, some studies proposed stress relaxation schemes. Chen et al. [92] detected stress by using heart rate collected from Zephyr wearable sensor. After the detection of stress, they created a user-specific breathing pattern which helps a subject to relax. The breathing patterns consist of three phases: inhalation, hold and exhalation. To restore the respiratory pattern of the user to a non-stressful pattern, they offered users a YOGA respiratory pattern which is the most similar to the user's respiratory pattern. They stated that following the YOGA respiratory patterns help users to relax.

Akmandor et al. [62] developed a stress detection and alleviation system by gathering ECG, GSR, respiration, blood pressure and blood oximeter in the laboratory environment. A memory game, fly sound, ice test and IAPS were used to induce stress. For alleviation, classical music, warm stone and good news were used. SVM and kNN were applied for 2-class classification. They achieved 95.8% accuracy. According to the results, alleviation methods make people return to the relaxed state faster.

In [86], researchers designed an experiment in a laboratory environment with 9 participants. They employed both the wrist-worn and chest based heart rate sensors and wrist-worn and finger-based GSR sensors. They classified the experiment into stressor tasks (ice bucket, singing, scary game, SCWT and arithmetic task) and non-stressor tasks (social conversation, eating, email reading and answering and homework). They first investigated the contribution of the wrist-worn and chest-based heart rate sensors and concluded that they did not affect the f-measure significantly (only 1%) They further tried to detect the task type with physiological sensors and obtained approximately 60% f-measure with the logistic regression, SVM and Random Forest classifiers. When they used the self-report as the stress level, they achieved between 60–66% accuracies with LR and SVM and 79% f-measure with the random forest. They concluded that stressor tasks have different effects on individuals and an increase in f-measure when self-reports are used as the ground truth is expected. They further planned to extend this experiment to daily life environments.

7.1.9. Insights from laboratory environment experiments

The laboratory environment experiments can be counted as the

Table 4
Stress detection experiments in controlled laboratory environments.

Article	Stress signal	Stress test	Method	# of Classes	Accuracy %
[50] (2012)	HRV	Stress in the traffic	Minimum Distance Classifier	3 (Low, Medium, High Stress)	90
[51] (2018)	HRV	SCWT	SVM	2 (Rest (R), Stress (S))	70
[67] (2010)	EDA – PPG	Hyperventilation and Talk Prep.	kNN – FDA	2 (S,R)	95
[54] (2010)	EDA	MIST	LDA – SVM-RBF – NCC	2 (Stress, Cognitive Load)	82.8
[66] (2011) (Extension of [67])	EDA – PPG	Hyperventilation and Talk Prep.	Fuzzy Logic	2 (S, R)	99
[64] (2015)	EDA – PPG	TSST	SVM	2 (S,R)	80
[70] (2013)	Speech	TSST	SVM	2 (S,R)	72
[65] (2011)	ECG- EMG – EDA	Arithmetic, Puzzle, Memory Tasks	Bayes, kNN, Fisher Least Square	2 (S, R)	80
[71] (2016)	PPG – EDA – Respiration – Thermal Cam	Lie Detection	DecisionTree	2 (S, R)	73
[72] (2016)	EEG	Arithmetic Task	SVM	4 (Neutral, Medium, Low, High Stress)	89
[73] (2015)	Body Movements	Arithmetic Task	SVM	2 (S, R)	77
[74] (2016) (Extension of [73])	Body Movements – EMG – EDA – Respiration	Arithmetic Task	SVM	2 (Stress, Relax)	85
[75] (2017)	Facial Cues	Social Exposure and Stressful Media (IAPS)	kNN – SVM – Naive Bayes	3 (Neutral, Relax, Stressed)	91.68
[76] (2011)	Pupil Diameter	SCWT	FuzzySVM	2 (S, R)	70
[77] (2014)	Pupil Diameter	IAPS	DecisionTree	2 (Stress, Relaxed)	90
[61] (2017)	EDA – PPG – Speech – Accelerometer	TSST	Adaboost	2 (Stress, Relax)	94
[55] (2015)	EDA – Accelerometer – Bluetooth	–	Logistic Regression	2 (Stressed, Unstressed)	91
[63] (2017)	HRV – EDA	Puzzles	F-state Machine	3 (Low, High stress, High alert)	0.984 (f-measure)
[78] (2012)	Temperature – Heat Flux – EDA- Respiration – Accelerometer	Arithmetic Task, Cold Pressor and Noisy Sounds	Naive Bayes	2 (Stress, Relaxed)	82
[62] (2017)	ECG – GSR – respiration – Blood Pressure – Blood Oximeter	Ice test and IAPS	SVM – kNN	2 (Stressed, Relax)	95.8
[79] (2014)	Added unsupervised activity clustering to [78]	Arithmetic Task, Cold Pressor and Noisy Sounds	Naive Bayes – Logistic Regression	2	65
[80,81] (2017) (Extension of [80])	Respiration	SCWT – Mental Computation	CNN	2 (Stressed, Relax)	84.59
[82] (2014)	EEG – ECG – EMG – EOG	Mental and Memory Task	ANN	3 (Relaxed, Mental, Fatigue Stress)	80
[83] (2015)	Facial Blood Flow	SCWT	Multiple Regression	2 (S, R)	88.6
[84] (2014)	Facial Blood Flow	TSST	Binary Classifier	2 (S, R)	88
[16] (2015)	EDA	Fail Scenarios	LDA	2 (S, R)	98.88
[85] (2016)	Human Gaze – Mouse Click	Arithmetic Task	Random Forest	2 (S, R)	66
[86] (2017)	EDA – PPG	Ice Bucket, SCWT, Mental Arithmetic Task, Singing	SVM, Logistic Regression, Random Forest	2 (S, R)	0.79 (f-measure)

Table 5
Camera based stress detection schemes.

Article	Stress signal	Stress test	Method	# of Classes	Accuracy %
[73] (2015)	Body Movements	Arithmetic Task	SVM	2 (S, R)	77
[74] (2016) (Extension of [73])	Movements – EMG – EDA – Respiration	Arithmetic Task	SVM	2 (Stress, Relax)	85
[75] (2017)	Facial Cues	SCWT	kNN – SVM – Naive Bayes	3 (Neutral, Relax, Stressed)	91.68
[76] (2011)	Pupil Diameter	SCWT	FuzzySVM	2 (Stressed, Relaxed)	70
[77] (2014)	Pupil Diameter	IAPS	DecisionTree	2 (Stressed, Relaxed)	90
[80] (2017) and [81] (2017) (Extension of [80])	Respiration	SCWT – Mental Computation	CNN	2 (Stressed, Relaxed)	84.59
[83] (2015)	Facial Blood Flow	SCWT	Multiple Regression Equation	2 (stressed, relaxed)	88.6
[84] (2014)	Facial Blood Flow	TSST	Binary Classifier	2 (Stressed, Relax)	88
[85] (2016)	Human Gaze – Mouse Click	Arithmetic Task	Random Forest	2 (S, R)	66

preliminary works for the stress detection research in the daily life. When we examine Table 4, we can deduce that EDA and HR combination yields the best results in the laboratory environments. Four out of five works, with the accuracies higher than 95%, use this combination as the physiological signals. LDA, SVM, kNN and Fuzzy Logic classifiers have achieved the highest accuracies with these signals (SVM was applied in [64] which is a more recent paper). In [72], researchers achieved 89% accuracy in 4-class stress classification by using EEG signals. However, for daily life stress detection, EEG measuring devices are obtrusive for individuals and they would not be comfortable wearing them in their daily life routines. Almost all of the studies in Table 4, employed a 2-class classification. To get more realistic results, stress detection resolution must be augmented (i.e., the number of stress levels should be increased). The laboratory experiments further demonstrated the importance of using multimodality for increasing the detection accuracies as in the case of EDA – HR combination.

7.2. Restricted environments

7.2.1. Office Environment (see Table 6)

One of the first stress detection works in the office environments was conducted by Hernandez et al. [93]. They investigated stress levels of the call center employees. They used skin conductance signals from an Affectiva QTM sensor for this purpose. They also evaluated the self-reports and call logs to detect stressful and non-stressful calls. For one week, nine employees answered 1500 calls. They altered the loss function of SVM to react better to changing priors when classifying calls. They achieved 78% classification accuracy for individuals and 73% for the general model.

Arnrich et al. [47] investigated the relationship between the posture changes and stress levels. The Montreal Imaging Stress Task was applied to induce stress. They investigated stress by comparing with a control condition where subjects have to perform the same task without a time pressure and without social stress. They calculated the Center of Pressure (CoP) from 1024 pressure sensor elements. To discriminate between cognitive load and stress, Self Organizing Maps (SOM) and XY-fused Kohonen Network were used. They observed that the variance in the sidewards is increasing in the stress condition. They achieved 73% classification accuracy when discriminating between stress and cognitive load.

Garcia-Ceja et al. [94] detected stress using only the accelerometer sensor of a smartphone. The advantages of the accelerometers are low privacy concerns, low power consumption, widespread usage in electronic devices such as smartphones, smart bands, fitness trackers. The Oldenburg burnout inventory (OLBI) questionnaire was used to collect reported stress data from a subject three times a day. They extracted 34 features in both the time and frequency domains. Naive Bayes and decision tree classifiers are used. They measured their classification accuracies in three different ways. The first one is user-specific results which have the highest accuracies, but they need more labeled data. The second model is the general model which has the lowest accuracy as expected. The novelty in this paper is the development of the third model which is the similar users model. Two users are selected similar by using some of their data and k-means clustering. Then, the model was applied to these users. The accuracy results (60%) are between the user-specific and the general model values. As future work, instead of collecting data all the time, only phone handling times can be used for data collection.

Kocielnik et al. [96] developed a stress detection scheme for a work environment. A DTI-2 sensor wristband was used to collect the EDA data. They further collected calendar information and the Self-Assessment Manikin Questionnaire user feedback. They preprocessed the data by removing the signal from the first 15 min and last 10 s, moments when the band lost contact with the skin and smoothing the signal. They did not provide classification accuracy, but they stated that the data is meaningful and provides new information.

Cinaz et al. [95] investigated workload levels in an office scenario. They used subjective ratings from NASA-TLX. They also had an objective reference from collected salivary cortisol samples. Zephyr Bioharness was employed to gather ECG data. They had a calibration phase where the subject's responses were measured. They defined three tasks with low, medium and high workload for the calibration. After the calibration phase, subjects are asked to perform their office works for one hour. Time and frequency domain features were used. They measured the mental load of office work and classified as low, medium or high. The LDA was the best performing classifier. They correctly classified mental workloads of 6 out of 7 subjects when subjective ratings were taken into the consideration.

Insights from the Experiments in Office Environment. Office environments worked as a bridge in the transition from stress detection in the

Table 6
Stress detection experiments in office environments.

Article	Stress signal	Method	# of Classes	Accuracy %
[93] (2011)	EDA	SVM	2 (Stressed, Relaxed)	78
[47] (2010)	Posture Changes	SOM – XY-fused Kohonen Network	2 (Stress, Cognitive Load)	73
[94] (2016)	Accelerometer	Naive Bayes – DecisionTree	3 (Low, medium, high stress)	60
[95] (2013)	ECG	LDA	3 (Low, Medium, High load)	85

Table 7
Stress detection experiments in automobile environments.

Article	Stress signal	Method	# of Classes	Accuracy %
[97] (2017)	PPG – IMU	SVM-RBF	2 (Stressed, Relax)	95
[98] (2016)	IMU	SVM	2 (Stressed, Relax)	94
[100] (2015)	ECG	SVM-RBF – kNN – RBF	2 (Stressed, Relax)	83
[101] (2015)	EDA – EMG – ECG	kNN – SVM	3 (Low, medium, High stress)	98
[102] (2015)	ECG	DecisionTree	3 (Low, medium, High stress)	88
[99] (2017)	ECG – EDA – Respiration	SVM-ELM	3	99
[103] (2013)	HRV	kNN	2 (Stressed, Relax)	97

laboratory environments to the real-life stress detection research. The movement of individuals is limited in the offices and the environment can be controlled with cameras and sensors. Having said that, offices and other workplaces are among places which contribute to stress levels of individuals. When the research in office environments is investigated (Table 6), we can see that works that employed EDA and HR have the highest accuracies. SVM and LDA were the best-performing classifiers.

7.2.2. Automobile environment

Stress in the automobile environments has been another interesting topic for researchers (see Table 7). Lee et al. [97] developed a stress detection system for drivers by using a wearable glove which has a photoplethysmogram (PPG) and inertial motion unit (IMU) orientation sensor which has an accelerometer, gyroscope and magnetometer on it. They gathered data from twenty-eight subjects and used Euro Truck driving simulator with a throttle pedal, brake pedal and steering wheel. A driving behavior survey (DBS) was filled by subjects. They rated anxious driving behavior using Likert type scale 1–7. They measured cardiac rhythms (PRV) using PPG. For both sensors, time and frequency domain features were extracted and an SVM classifier was applied. The best result for 2-class classification was 95% using SFS-SVM with an RBF kernel function (Table 7).

Lee et al. [98] proposed a driver stress detection system using only an inertial motion unit. Their experimental setup was the same as [97]. As the stress level references, they used GSR, self-report surveys and facial expressions. Eight subjects participated in the experiments. They extracted 46 features and reduced to 22 with an insignificant loss in the classification accuracy. By using SVM classifiers, they achieved approximately 94% accuracy for discriminating low stress from high stress. As future work, they plan to investigate which conditions cause an increase in the driver's stress.

Chen et al. [99] detected driver stress by evaluating ECG, EDA and respiration data. Time, frequency and wavelet domain features were extracted. PCA and Sparse Bayesian Learning (SBL) machine learning algorithms were applied. They achieved 99% accuracy on three class classification.

Munla et al. [100] proposed a driver stress detection system. They used the Automobile Driver database (DRIVEDB) [104] including ECG signals collected from 16 different individuals driving around Boston, Massachusetts. From the data, they extracted time domain, frequency domain, nonlinear and time-frequency domain (such as wavelet and STFT) features. As classifiers, they used SVM-RBF, kNN and RBF. They achieved 83% classification accuracy when discriminating between stress and no stress conditions. They plan to extend their work into distinguishing more stress levels.

Ghaderi et al. [101] presented a stress detection scheme for automobile drivers. They used DRIVEDB from PHYSIONET database created by Jennifer Healey and Rosalind Picard [105]. The data were recorded from drivers in Boston. They used GSR, EMG, ECG signals. They selected reported best features from the literature for this signals. For 100 s intervals, they classified the driver stress as low, medium and high. By using kNN and SVM, they achieved 98% classification accuracy over this database. They stated that respiration is the most discriminative signal to detect stress.

Another research based on DriveDB database was carried out by Keshan et al. [102]. They only used ECG data from the database. They stated that their difference was to develop personalized stress analysis consisting of three levels of stress: low, medium and high. ECG has become less invasive with wearable patches and demonstrates a high correlation with stress. They used Naive Bayes, Logistic Regression, Multilayer Perceptron, SVM, J48, kNN and random forest classifiers. They used extracted features alone and in groups. The decision tree gave the best results in most cases. The single feature that discriminates stress solely best was the difference in the average number of beats. They achieved 88% classification accuracy with three classes of stress.

DriveDB database was used in [103] to measure driver stress levels. Wang et al. extracted features by employing trend-based and parameter-based methods. In trend-based methods, long-term statistical features were calculated. Parameter-based methods extracted features from 5-min HRV analysis. The kernel based class separability (KBCS) technique was employed to select features. LDA and PCA were used to reduce feature dimensions. LDA, PCA and KBCS were used in different combinations. They showed that the best accuracy was achieved when all three methods were used with trend-based features. Long-term features are more distinctive than parameter based short-term features. They reached 97% classification accuracy by using a kNN classifier.

Stress levels of bus drivers were investigated in [17]. Rodrigues et al. developed a bus driver stress detection scheme. They used the VitalJacket biodevice to measure ECG. As a reference, they used pressing the button for stressful events and at the end of the day, a questionnaire filled by bus drivers. They gave the extra information and visualization using Google Map and GPS data to help the driver remember the stressful event. They collected data from 36 drivers. Another significant result is that they found stressful locations in Porto, Portugal which have tight roads, low visibility crossroads, specific intersections. This information can be helpful to municipalities to alleviate problems in these areas. The LF component in HRV was the most discriminative feature. They also demonstrated that experience and stress levels are inversely proportional by using the background data from drivers.

Insights from the Experiments in Automobile Environments. In the automobile environment, movements are restricted and sensors like cameras can be integrated into the ambient. In that sense, this environment is similar to the offices and workplaces. People tend to be stressed in automobile environments especially in crowded cities with traffic jams. In the literature, most works have used the driveDB database [105]. Fifty minutes ECG, EMG, EDA, respiratory sensor data was collected from 24 drivers in Boston in this database. EDA and ECG signal combination has achieved the best two performances in the automobile environment. For the driveDB database, SVM and kNN classifiers achieved the best accuracies (up to 99% [99]) in a 3-class classification. Another significant research in this environment is the bus driver stress detection in Porto [17]. They stated that stress has a negative impact on memory and this affects the accuracy of the ground truth questionnaires. They added photos of stressful events for drivers to help them remember the important times at the end of the day. They further demonstrated that the LF component in the HRV is the most distinctive feature.

7.3. Semi-restricted environments

7.3.1. University campus environments: student stress

Detecting stress in university campus environments would help students to increase their academic success and quality of their lives. Wang et al. [110] conducted a very wide range research on college students. They developed a studentLife application which collects activity, sleep, conversation data and EMA (Ecological Momentary Assessment) questionnaires filled by participants. Their aim was to evaluate mental well-being, stress, loneliness and their correlation with academic performance. They made the dataset publicly available. They showed that sociability, location changes, activity of students decreases when a midterm approaches. They found a correlation between GPA and conversation and indoor mobility of students. Although the research is not about stress detection, the data collected from students can be used to recognize stress.

Castaldo et al. [106] investigated student stress during an oral exam by using ECG. The difference from other HRV using articles is that they detected stress using ultra short-term HRV (3 min). They recorded controlled resting base condition from the same student after a vacation. 18 features were extracted from the time and frequency domains. Non-linear features were also used. 12 out of 18 features correlate with stress. Naive Bayes, Decision Tree, SVM and Multilayer Perceptron classifiers were applied to the data. They achieved the best accuracy with the C4.5 tree classifier which is approximately 80%. The strong sides can be listed as the measurement of daily life stress of students during an oral exam and ultra short-term HRV usage.

In earlier research, Gjoreski et al. [107] developed a student stress detection scheme by using the data from smartphones which was collected during the study in [110]. They used the activity (from the accelerometer), audio classification (silence, voice, noise), GPS, WiFi, conversation (whether there is one nearby), time and duration of calls, light sensor data and questionnaire data from subjects. 47 features were extracted from these data. They applied three different approaches. The first is the general model for all subjects with Leave One Student Out testing, they obtained 43% classification accuracy when discriminating not stressed, slightly stressed and stressed cases. They divided students who react similarly into clusters and calculated inside cluster accuracy, but results are the same with the general model. The last classification was dividing data of each student into two parts. The first part of the data was used as a training and rest of the student data were used as the test data. They called this method learning with a calibration phase and achieved 60% classification accuracy. They stated that low classification accuracy caused by subjectiveness of the perceived stress and difficulty of determining stress using only smartphones. For future works, they plan to add voice analysis from smartphone and social media analysis.

Bogomolov et al. [111,108] proposed a scheme that automatically measures the stress of graduate students. They collected data from 117 subjects for approximately a year. The selected features were the user's mobile phone activity (call and SMS logs, Bluetooth proximity hits), weather conditions, personality traits. They divided weather condition into six: mean temperature, pressure, total precipitation, humidity, visibility and wind speed namely. They also classified personality traits into the Big five: extraversion, neuroticism, agreeableness, conscientiousness and openness to experience. They decided personality by applying Big five questionnaire. The stress reference was determined from self-perceived stress level questionnaire filled at the end of a day. As a classifier, a number of classification algorithms were applied, but random forest outperformed other algorithms. They reduced 500 features into 32 features by applying the Pearson correlation analysis. They achieved 72% classification accuracy. The weaknesses of this research can be counted as a biased sample (all grad students) and interaction with people who are not part of the study was not taken into account. Their final aim is to integrate the non-obtrusive system into the clinical support or office self-monitoring applications.

Bauer et al. [109] presented a mobile phone stress detection scheme. They defined a two week exam period as a stressful time and after an exam a two week period as a stress-free time on participating students who were given an Android mobile phone with the required logging software. They divided behavioral patterns into three classes: location behavior, social interaction behavior, call and SMS behavior. They defined Regions of Interests (ROI) and tracked it with GPS and Wi-Fi. The k-means algorithm was employed to determine these regions. They measured social interaction with the number of different Bluetooth device connections. They measured the difference in behavior between the exam period and the free period. They showed that participants changed their behavior 53% on average. For future works, they plan to add new behavioral features (outgoing SMS, phone call duration, etc.), detect behavior change in real time and add a mobile questionnaire.

Insights from the Campus Environments. Campus environment is the most similar environment to the daily life experiments among the restricted environments. Movements are not limited and campus environments are less controllable with sensors when compared with offices and automobile environments. Therefore, the accuracy of the classification schemes is lower than these environments (See Table 8). The ECG signal and the decision tree classifier achieved the highest accuracy in 2-class classification. Most works only used features extracted from the smartphones. Wearables with more sensors (such as EDA, temperature, accelerometer) can be integrated into these systems to improve the performance of these systems.

7.4. Nonrestricted daily life

The ultimate aim of the studies is to detect stress levels of individuals in their daily lives (see Table 9). Data is collected with unobtrusive wearable devices (Fig. 7). The number of participants and data collection intervals are provided in Table 10. Ciman et al. [112] developed a stress detection scheme by analyzing the smartphone usage patterns. They divided the experiment into two parts. The first part took place in the controlled lab environment. They developed an Android application with search and write tasks. In these tasks, users tap, scroll, swipe and text input gestures were recorded. They provided a stressor task to induce stress on the subject. For reference, 5-point Likert scale stress states were reported using the Experience Sampling Method. There were 13 subjects for this part. They achieved approximately 80% stress detection classification accuracy using SVM, NN, kNN and decision tree classifiers. For the second part which they called "in the wild", they collected used application types, physical activity of the user, light values of screen and events related to a mobile phone screen. The subject is using the smartphone in his/her own daily life in this part. They achieved approximately 70% classification accuracy. The weakness of the second part in used application type evaluation is the issue of causality. They reported that they do not know whether the used application causes the stress or stress causes the used application type. For instance, a researcher might notice that when a subject is stressed, the used application type is generally social media. However, they could not conclude that social media app usage caused stress. Because another possibility might be that the increased stress level caused the social media app usage. They emphasized the causality issue in that case. As future works, they plan to build a remote assessment system for data collection and add wearable devices to the system.

Gjoreski et al. [113] designed a stress detection scheme in both daily life and laboratory. They used an Empatica wrist device. The mental arithmetic task was used to induce stress. From BVP, HR, ST, GSR and RR sensors, 63 features were extracted. In the laboratory environment, they discriminate stress from no stress with 83% accuracy. They also introduced a three-class classifier with no stress, low stress and high stress with 72% classification accuracy. However, for the daily life, they used activity recognition which discriminates sitting, walking, running and cycling. The activities are enumerated according to their

Table 8
Stress detection experiments in university campus environments.

Article	Stress signal	Method	# of Classes	Accuracy %
[106] (2016)	ECG (Ultra Short Term Features)	SVM – DecisionTree – NaiveBayes – MultilayerPerceptron	2 (Stressed, Relax)	80
[107] (2015)	Accelerometer – Speech – GPS – Wifi – Proximity – Call Statistics – Light Sensor	SVM – J48 – Bagging – RandomForest – Ordinal classifier	3 (Not stressed, Slightly stressed, Stressed)	60
[108] (2014)	Call-SMS Statistics – Bluetooth – Weather Conditions – Personality Traits	Random Forest	2 (Stressed, Relax)	72
[109] (2012)	GPS – Social Interaction (Bluetooth Connections) – Call-SMS Statistics	–	2 (Stressed, Relax)	53

Table 9
Stress detection experiments in unrestricted daily life.

Article	Stress signal	Method	# of Classes	Accuracy %
[112] (2016)	Mobile Application Usage Pattern – Physical Activity – Light Sensor – Screen Events	SVM, ANN, kNN	2 (Stressed, Relax)	70
[113] (2016)	BVP – SkinTemperature – EDA – RR – HeartRate (Without Context Info)	RandomForest	2 (Stressed, Relax)	76 (With Context Information 92)
[114] (2015)	Speech – Gyroscope – Accelerometer – Light Sensor – Screen Events – Activity Type	RandomForest – Simple Logic – DecisionTree	2 (Stressed, Relax)	77
[115] (2014)	ECG – SkinTemperature – Respiration – Accelerometer- EDA	SVM – kNN – ANN – RandomForest	2 (Stressed, Relax)	73
[116] (2013)	Call-SMS Statistics – GPS – Screen On/Off – Accelerometer – EDA	SVM – SVM-RBF – kNN	2 (Stressed, Relax)	75
[117] (2013)	HRV – Speech – Physical Activity – GPS – Accelerometer	Logistic Regression	3-class (Low, moderate, high stress)	61
[118] (2015)	ECG + Respiratory + Accelerometer	SVM	2 class (Stressed, Relax)	72
[15] (2011)	ECG + Respiratory	J48, J48 + Adaboost, SVM, (HMM for daily life)	2 class (Stressed, Relax)	0.71 (correlation with self-reports)
[119] (2012)	Speech	GMM	2 (Stressed, Relax)	80.5
[120] (2017)	HR – IBI – HRV – EDA – Temperature	Weka Toolkit	2 class (Stressed, Relax)	70 (precision with 95% recall)
[121] (2018)	Usage Data for different application categories	HMM with MPM	2 (Stressed, Relax)	68



Fig. 7. Wearable Devices for collecting data in Daily Life (Left Top: Fitbit, Right Top: Sony Smartband, Middle: Samsung Galaxy Gear S3, Bottom: Empatica E4).

intensity (i.e lying corresponds to 1 to running corresponds to 5). For the stress detection intervals, the average intensity is calculated in this manner. The reason behind activity recognizer and intensity calculation is to distinguish an intense physical activity from a stressful situation by giving context information to the daily life stress detector as an input. They divide a day into one-hour episodes. The subjects record stressful events by pushing a button. They achieved 76% and 92% stress detection classification accuracies for no-context and with context information (using activity recognizer) cases respectively. They used 11 days of daily life data from 5 users. As future works, they plan to discriminate more stress levels also. Personalization depending on age, gender, fitness condition is also another planned future work. They expanded their work in [120]by creating a stress detection model for the laboratory environment (2-min interval), activity recognizer and context information. In the laboratory, they recorded a one-day relax baseline and mental arithmetic task as the stress phase. They mapped the subjective scores to stress labels in this phase. Daily life system uses these models and decides for every 20 min. From 5 participants, daily life data collected for 11 days. HR, BVP, IBI, EDA and temperature data were obtained. They applied all ML algorithms in the MATLAB Weka toolkit. They achieved 70% recall rate with 95% precision.

Gimpel et al. [122] presented a stress detection scheme by using the smartphone data. They stated that their most significant innovation from similar works is that their system is not based on user inputs or additional devices. They published an Android app and analyzed the data from users. They made use of 36 hardware and software sensors. Their research was in progress; thus they did not provide results. However, they found out that high smartphone usage, average battery temperature, the maximum number of running applications and the frequency of switching the display on are related to high stress. They also pointed out that in daily life, we do not know whether high smartphone usage causes stress or vice versa. Furthermore, they reported that the perceived stress is not necessarily the same as the actual stress.

Another daily life stress detection scheme using smartphones was presented in [114]. Sysoev used audio, gyroscope, accelerometer, ambient light sensor data, screen mode changing frequency, self-assessment and activity type. They did not use any wearable sensors. Self-assessment stress level based seven scale questionnaire NASA-TLX was used as a reference. They combined activity recognition with their stress detection system to increase its performance. They achieved approximately 77% classification accuracy which corresponds to 3.8% increase when compared to the stress detection scheme not using the activity recognizer. They created a separate model for the standing activity among other which results in 1.5% accuracy increase.

Maier et al. [123] presented their work in progress. They used HRV with activity and contextual data to increase the accuracy of the stress detection scheme which uses ECG. They monitored the stress level of a user and when the predetermined threshold was exceeded. Users can quit from the stressful event or some relaxation methods are provided to them when they are stressed. Their app has an automatic adaptation for individuals. The app collects data from the user about emotion and energy. Physical activity from the accelerometer, the location from the GPS, change of location and time of the day information were added to HRV to increase the accuracy. They applied a particular neural network classifier (BINN). However, they did not provide its accuracy. They plan to adapt the work for mentally ill people in addition to healthy people. They also plan to add the Perceived Stress Questionnaire (PSQ) as a reference.

Muaremi et al. [115] presented a system for detecting stress from the sleep patterns. They collected data from 10 Hajj pilgrims. Both physical and physiological data measurements are obtained. ECG/HRV, respiration, body temperature, GSR, upper body posture sensors, accelerometers on body and arms were the data sources. Chest strap and Empatica wristband were used to collect data. After features were extracted, SVM, kNN, NN, RF (Random Forest) and logistic regression classifiers were applied to the data. SVM obtained the best classification accuracy. ECG/HRV, upper body activity and sleep duration were the most distinctive modalities. 73% maximum classification accuracy was achieved. For future works, in addition to the existing features, sleep stages, the number of wake-ups and percentage of time that the subject is lying but not sleeping could be taken into consideration.

Sano et al. [116] developed a stress recognition scheme using a wrist sensor, a mobile phone and surveys. Mobile phone usage was

Table 10
Campus environment and daily life experiment details.

Article	Devices	# of Participants	Experiment duration
[121] (2018)	Smart Phone	28	4 days
[124] (2017)	Smartphone	5	4 weeks
[106] (2016)	3-lead ECG	42	2 days
[112] (2016)	Smartphone	25	4 weeks
[113] (2016)	Smart Wrist (Empatica E4)	5	55 days
[123] (2015)	Smart Wrist (HRV data only)	35	4 weeks
[118] (2015)	ECG + Respiratory + Accelerometer (Wearable Sensors)	23 (3 good EMA reports)	1 week
[110] (2014)	Smartphone	48	10 weeks
[111] (2014)	Smartphone	111	7 month
[116] (2013)	Smartphone + Smart Wrist	18	5 days
[117] (2013)	Smartphone + Smart Wrist (HRV)	35	4 weeks
[109] (2012)	Smartphone	7	4 weeks
[15] (2012)	ECG + Respiratory (Wearable Sensors)	17	2 days

modeled from a call, SMS, location change (mobility) and screen on/off features. Accelerometer and EDA data were collected from a wrist sensor. They obtained information about sleep, caffeine and alcohol usage, stress, mood and tiredness from the surveys. Sequential Forward Floating Selection (SFFS) was employed to find the most discriminative features. SVM with linear and RBF kernels and kNN were applied for classification. PCA was also employed with classifiers. Acceleration during the second sleep phase and between 6 and 9 pm, few or short SMS and a small number of screen display on/off behavior between 6–9 pm or 9–12 am were found to be correlated with high stress. By employing screen on/off, mobility, call, acceleration and EDA data, they achieved 75% classification accuracy in daily life with two classes.

Kostopoulos et al. [124] proposed a stress detection application for the Android operating system called StayActive. They employed sleeping patterns, social interaction and physical activity to detect stress. They used a combination of offline mathematical model and an online machine learning algorithm to detect stress and suggested some relaxation activities when stress is recognized. They collected user stress feedback by using the modified Circumplex Model of Affect by Russel [125]. They added a relaxed-stressed question to this questionnaire. They used the sleeping hours as a feature for the sleep patterns. A punishment was given to lack of sleep and excessive sleep. They measured the social interaction by using the number of touches to screen, number of calls and SMSs. To compute feature weights, they asked the user how important they consider these features. These were the preliminary results. They plan to extend the number of people and add some physiological signals to the study as future works.

Muaremi et al. [117] presented a stress detection scheme for iOS devices. They gathered data from both smartphones and wearable chest belts. HRV data were collected from the chest belts and audio, physical activity, communication data were gathered from smartphones. They divided the day into four quarters and users were asked to fill out the questionnaire and record their voice in each quarter. This questionnaire was a modified version of the Positive and Negative Affect Schedule (PANAS) questionnaire. They removed some questions because the subjects complained about the length of the questionnaire. Before going to sleep, a subject provides stress level information for the whole day. They anonymized the speech and used the openSMILE library for the audio feature extraction. Physical activity data was gathered using GPS and accelerometer. Social interaction features were call events, number of calls, duration and ratio of incoming and outgoing calls. For the HRV data, time, frequency domain and nonlinear features were extracted. They modeled and scored daily and long-term stress for each individual. The stress model is dynamically changing in time. They removed highly correlated features and obtained 13 smartphone and 10 HRV features. After the feature selection, they noticed that accelerometer, calendar, battery features were not selected because subjects are not using the smartphones as a calendar, they used their personal computers for that purpose. Another reason was that they put smartphones on the table and activity data becomes useless. They further stated that the HRV data is more distinctive than the smartphone features. By employing logistic regression models, they achieved 61% classification accuracy for three class (low, medium and high stress) classification with all features. As future works, they plan to add comparative questionnaires with past time, apply SVM and random forest classifiers, develop a scheme that has sleep stage awareness and measure HRV during a day in addition to the sleep times.

In [118], researchers developed a stress detection scheme for continuous monitoring. They used a chest band as a respiratory sensor, a 2-lead ECG and a 3-axis accelerometer for data collection. They first designed a laboratory experiment with a baseline, public speech, arithmetic test and ice cold test sessions which are used to train a model with 24 (training)-26 (test) participants. They further collected field data from 23 participants in the wild. Participants are required to fill 15 questionnaires. They used the adapted version of the PSS survey which includes five questions. The SVM classifier was applied. In Laboratory testing, they achieved 89% recall. On the other hand, they obtained 72% accuracy on the wild. As future works, they planned to increase the accuracy of the system by applying new data processing methods, develop a stress management scheme, add some context information such as GPS, light exposure and social interaction.

Lu et al. [119] detected stress from human voice gathered from smartphones. They separately evaluated indoor and outdoor environment cases. They created three experimental setups: a job interview, marketing jobs and a neutral task. They used GSR data as a reference. For each task, they divide audio into voice and non-voice parts. They implemented different algorithms for speaker segmentation for indoor and outdoor environments. For the outdoor environment, they employed two smartphones for speaker segmentation, one in the shoulder another in the waist. Used features were pitch (F0), spectral energy distribution, speaking rate, nonlinear TEO transformation results and MFCC. The classification was performed based on GMM and likelihood functions. They employed three models: a universal model, personalized model and individual-adapted model (by using MAP). They stated that pitch features and speaking rate are the most distinctive features for stress detection. General models achieved 69% (outdoor and indoor average) classification accuracy whereas personalized models achieved 80.5%. Supervised adapted models achieved 10% higher than the general, 2% lower accuracy than the personalized model.

Reimer et al. [126] described their pilot mobile system on stress recognition. They chose only HRV signal stating that EDA is not wearable which is not a true statement. They also added the context information like GPS, physical activity, time of a day and week. Their main novelty in this app is the automatic user calibration. The algorithm sets thresholds to discriminate three stress levels at the initial learning phase. 35 healthy subjects participated in the experiment for four weeks. The PSQ questionnaire was collected at the end of each week. They did not present the accuracy of the system.

Labeled data is an essential problem for daily life stress detection algorithms. In most of the studies, collecting labeled data is a bottleneck and available labeled data has the subjectivity problem. Vildjounaite et al. [121] proposed an unsupervised stress detection scheme. They used mobile phone usage data and applied HMM with MPM for classification. 28 people participated in the experiment with an average of 4 days. They achieved maximum 68% accuracy for semi-personal data.

In [15], the authors collected data from 21 subjects in the laboratory environment and gathered self-reports. They developed two stress recognition models. The first one is the physiological classifier which uses ECG and respiratory signals for detecting stress of an individual. They divide the experiment into the baseline, public speaking, mental arithmetic and cold pressor sessions. By applying J48, J48 + Adaboost and SVM classifiers, they obtained 90.17% accuracy in a two-class (stressed or non-stressed) classification task. They further developed a perceived stress model to estimate the stress perception. This model takes the outputs of a physiological classifier as an input and predicts the accumulation and decay of the stress level of a participant by employing HMM. They calibrate parameters of the model for each participant. This model has achieved 0.72 median correlation with self-reports in the laboratory environment. In a two day field study with 17 participants, the perceived stress model has 0.71 correlation with the self-reports. As future works, the researchers have planned to add more modalities, develop new methods to clean the signals during a physical activity and increase the number of the perceived levels of the stress.

7.4.1. Insights from daily life experiments

The ultimate aim of researchers is to detect stress levels and help individuals to cope with high stress in their daily lives. However, there are some issues to deal with when taking a step outside the laboratory (see Section 6). The stress detection accuracies of daily life schemes are lower than in restricted environments and laboratory environments. Smartphone usage statistics and wearable sensor stress detection schemes have accuracies between 70% and 80%. The combination of

smartphone usage data and physiological signals (such as PPG and EDA) from an unobtrusive wearable sensor data would increase the stress detection accuracy. One of the most significant issues in the daily life data collection is the ground truth collection and the reliability of the questionnaire data (see Section 2). Researchers developed an unsupervised stress detection for daily life by employing HMM in [15,121]. Unsupervised schemes may eliminate the need to collect the ground truth questionnaires. Gjoreski et al. [120] used activity recognition to have knowledge about context and improve their stress detection performance. Other context data such as information of whether the subject is inside/outside of a building by employing light sensors and GPS, calendar load, weekend/weekday knowledge, social interaction knowledge from Bluetooth sensors can be added to increase the performance of these schemes. The duration of experiments and number of participants are limited in the literature (See Table 10). To get a better understanding of stress behaviors and discriminative features, these numbers should be improved. Lastly, after detecting stress, some stress alleviation methods specific for individuals should be suggested to recover from the effects of stress. We will mention the scientifically proven stress alleviation methods in Section 8. We selected the methods that can be applied without interrupting daily routines and can be also used in indoor environments.

8. Stress alleviation methods: mobile apps and techniques

Another important question arises after detecting stress is how to relax an individual to the normal state i.e. emotion regulation. Some ancient techniques can be employed to alleviate stress such as yoga, meditation. However, these techniques either require outdoor environments and they could not be applied without interrupting daily activities or in office environments. We investigated mobile apps and techniques to alleviate stress and decrease its impact without leaving the office environments. The effect of these indoor techniques is not thoroughly investigated but a limited number of studies have been started such as [127].

8.1. Echo and emotical apps

Isaacs et al. [129] developed two mobile apps to cope with stress. The first one is based on users reflection on daily events. Users are asked to write reflections to daily events. After a while, both negative and positive reflections are showed to the user again and the user is requested to write another reflection on this event after some time. The reflection of negative and positive past both have their advantages. Negative past reflection reduced medical visits, improved immune response, better grades, reemployment, reduced absenteeism, increased subjective well-being and increased working memory [129]. Positive past reflection increases enjoyment of life, subjective happiness and it has a positive effect (often invoked as a response to loneliness) [129]. It causes escapism which is “A vacation from the present”. They evaluated these and developed a reflection mobile app “Echo”. The screenshots of the app are available in Fig. 8.

The second application from this research group is the Emotical app which aims to understand and change the mood of a person. Hollis et al. [130] listed the challenges of emotion regulation as people do not have much insight into their emotions, poor at predicting future emotions, poor at regulating mood by actions. They defined the problem and make complex inferences about what affects mood and then propose remedial actions. After the application detects high stress, it suggests some activities based on users information about what makes her/him happy in the beginning. It also clusters people and suggests joyful activities of people in the same cluster. This feature provides discoveries for users to learn what makes them happy.

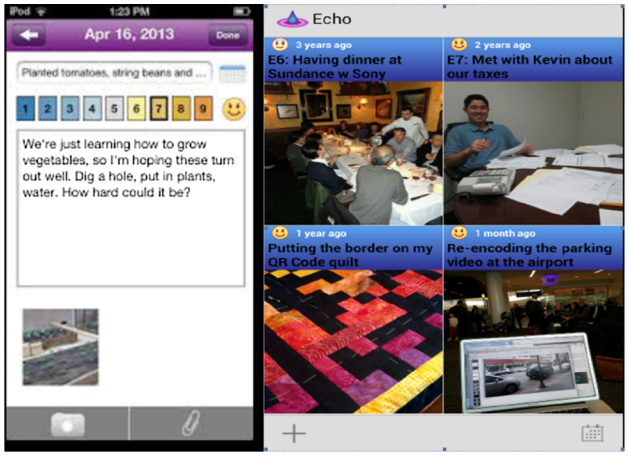


Fig. 8. Echo App for Reflections: Screenshots [128].

8.2. Pause and sway iPhone application – Tai-Chi on screen

Cheng et al. [127] indicated that humans have 65000 thoughts on an average day. Eighty percent of these are not positive and 95% of these thoughts are from the previous day. Their starting idea was paying attention at will increases the level of happiness [127]. They developed a mobile app inspired by an ancient Chinese martial art Tai Chi. They focused on the movement based medication. Pause app does not require any special equipment besides from a mobile phone and it could be used in busy, noisy daily situations [127]. Pause provides mindful touch on the smartphones and helps individuals relax by giving calm feedbacks. They stated that interactive meditation works better than traditional guided meditation in noisy daily life situations, meanwhile, in quiet environments, Pause has similar results with traditional techniques. A screenshot can be seen in Fig. 9:

8.3. HeartMath: increase your ‘Coherence’

HeartMath is a stress alleviation mobile application. Developers tried to increase inner balance which helps individuals to prevent, manage and reverse the harmful effects of stress. They give users some goals every day to achieve. Users breathe deeply and think of some positive memories. They are required to plug a HeartMath sensor to their phones and start the relaxation session. They will breathe along with the pacer. While breathing, they focus on a moment when they felt

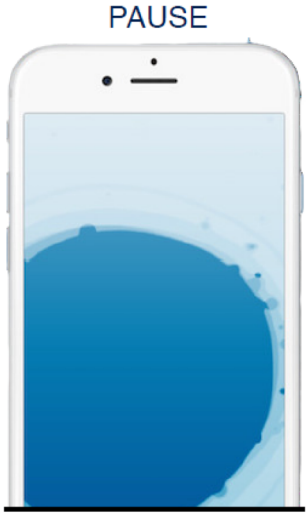


Fig. 9. You move your finger with the circle. If you move “mindfully”, the music continues and circle grows. Otherwise, feedback stops [131].



Fig. 10. Heartmath Screenshots: Real-Time Feedback [132].

joy, appreciation or care by closing their eyes. By real-time coaching, it shows them how coherent they are. Coherence is a formula that HeartMath created. If a user's heart signal is more like a sinusoidal, they are more coherent. They can increase their coherence and decrease their stress levels by the 5 min exercises. They can track their progress to keep the motivation up. Screenshots are given in Fig. 10.

8.4. Other relaxation techniques used in the literature

There are other apps in the literature. The Spire device measures breathe per minute, evaluates and if an individual is stressed, offer some respiration exercises [7]. Similarly, Tinke app evaluates a novel metric called the Zen index [7]. If a user is stressed, the Zen index will be higher. Tinke then suggests deep breathing exercises. Another app is WellBe. WellBe advises some breathing and meditation exercises that utilize sitting in silence and some relaxing voices [7]. Akmandor et al. [62] used classical music, warm stone and good news to relax a stressed individual. Chen et al. [92] first record the respiratory pattern of participants. To relax the stressed participant, they suggest a YOGA respiratory pattern which is most similar to the stressed user's respiratory pattern.

8.5. Insights from stress alleviation apps and techniques

An ideal stress alleviation method should be applicable indoor, scientifically proven and not require any extra instrument. Pause app is the closest one to the ideal stress alleviation technique definition. HeartMath is also a significant alleviation method. The disadvantage of HeartMath is the need for an extra equipment. Echo app is more suitable for dealing with chronic stress. Yoga is another effective stress reducing technique but it is hard to apply in daily life routines and in workplaces. Classical music, good news, warm stones may affect some people and not affect others.

9. Discussion and open challenges

In this section, we examined the open research problems. We further provided the insights from the literature. The studies that have the most important contributions are also discussed.

9.1. Discrimination of similar emotions

Because of the unresolved challenges of daily life emotion monitoring such as noisy data, many parameters that might induce stress, limitless movement of subjects, stress detection in daily life remains as an open challenge for researchers. The first problem and research challenge of daily life stress detection is caused by human physiology. For some physiological signals like HR and respiration, different emotions can result in similar signals [6]. Other emotions can activate SNS and generate a similar heart rate signal. Discrimination of these emotions is an open research problem.

9.2. Unknown context and conditions

Another research problem is related to non-restricted environments and stimuli in daily life. Since researchers cannot have all information about individuals all the time, different conditions such as increasing the room temperature, exercising and audio-visual stimuli can generate misleading signals. Context information can be collected and used to augment the performance of the systems [113].

9.3. Artifacts from unlimited movement

Unrestricted movement of people and improperly worn devices are other significant challenges. In laboratory environments, movements and activities are limited and constrained. Researchers have the chance to intervene with subjects to wear devices properly. However, in daily lives, movements are unrestricted. People even tend to make more than one activity at a time which makes the detection process more complicated. This could reduce the performance of stress detection systems and increase motion artifacts. Activity recognition schemes may be employed with stress detection systems to increase accuracy.

9.4. Subjective ground truth

The last open research problem is limited and unreliable labels. Since we do not have the ground truth in daily life, we rely on users' subjective reports, surveys and questionnaires as references. However, since these are subjective, they could direct the system in the wrong way and change from person to person. In the same situation and having similar physiological signals, two people can fill questionnaires as "stressed" and "non-stressed" or mark them at different levels [16]. This will reduce the performance of systems. Furthermore, label time may be marked incorrectly by users which diminishes the accuracy of stress detection schemes. User subjective score calibration can be done in a laboratory before the daily life experiment. In this way, we can learn the subjective score – level of stress in a situation mapping and evaluate the subjective ratings accordingly.

9.5. Power consumption

Especially for the daily life experiments, the battery life of the wearables becomes important. Off-the-shelf commercial smartwatches have 3–4 h of battery life when all the sensors are active. In order to collect a whole day of data, the power consumption of the devices should be decreased. To this end, the duty cycles of the employed sensors can be changed without significantly affecting the accuracies of stress detection systems. Open source operating systems could also be modified (i.e. closing some background processes, unused default programs). The power consumption aspect is not investigated in the literature. The ideal technique would increase the battery life of the devices during stress monitoring without decreasing the detection accuracy much.

9.6. Insights from the literature

The most distinctive and unobtrusive signals for detecting stress are EDA and HR. Especially frequency domain features such as LF, HF and LF/HF in the heart activity are mentioned as the distinctive features in the literature. From the smartphones, screen on-off frequency, SMS and call logs are the most discriminative features. The combination of these

features from smartphones and wearable devices (EDA, HR sensors) would be necessary for developing a successful stress detection scheme. Unsupervised HMM models for detecting stress in daily life have been employed in the literature which may remove the need for collecting the ground truth surveys. Collecting context information (activity, social interaction, GPS and environmental light) about a user will also help researchers to anticipate the state of individuals. Furthermore, detecting stress of an individual is not enough. Researchers have to offer some relaxation activities to return them to their relaxed state.

10. Conclusion

Stress is an important challenge in the modern society and it has social and economic effects. Research has started in controlled laboratory environments and these works have high accuracy for detecting stress. However, since daily life stress level can be significantly different from stress level induced in laboratory environments [1], daily life experiments have become popular in the research community. Another essential reason for daily life stress detection studies to be more attractive for researchers is the fact that users do not want obtrusive measurement techniques used in laboratory environments. Smartphones and unobtrusive wearable devices can be used to measure stress levels in daily life without bothering the users. In this section, we discussed open research challenges for daily life stress detection and there is a room for improvement in this area. Furthermore, the accuracies of daily life stress detection schemes are also much lower than those of the laboratory environments. The ultimate objective in stress detection is to develop a high accuracy scheme in daily life by overcoming unresolved challenges and applying emotion regulation methods to alleviate the stress of the users.

Conflict of interest

The authors declared that there is no conflict of interest.

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